

Short title: FAT Mutations and Resistance

**Biochemical Characterization of Fatty Acid Thioesterase Target Site Mutants and their
Implication on Herbicide Resistance**

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Abstract

Herbicide resistance threatens effective weed control in modern agriculture, particularly in grass weeds such as *Alopecurus myosuroides* and *Lolium multiflorum*. Cinmethylin is a pre-emergence herbicide with a novel mode of action that inhibits plastidial fatty acid thioesterases (FATs), enzymes essential for fatty acid biosynthesis. Although no cases of field resistance to cinmethylin have been reported, its resistance risk has not been fully assessed. In this study, we biochemically characterized defined amino acid substitutions in FAT A and FAT B to evaluate their effects on cinmethylin inhibition profile. Some substitutions in FAT A reduced inhibition *in vitro*, with mutations at residue R171 causing the largest shifts in sensitivity. However, these highly resistant variants required multiple specific nucleotide polymorphisms and are therefore predicted to be unlikely to arise in weed populations. In FAT B, sensitivity shifts were generally moderate. Importantly, most substitutions that reduced cinmethylin sensitivity also impaired enzymatic activity, suggesting limited viability *in planta*. Overall, these results indicate that while theoretical target-site resistance mechanisms exist, the practical risk of rapid resistance evolution to cinmethylin is low, supporting its value for integrated grass weed management

Keywords: amino acid substitution, blackgrass, cinmethylin, fatty acid thioesterase, herbicide resistance, Italian ryegrass, resistance evolution

Introduction

Since the domestication of crops, humans have faced the persistent challenge of protecting agricultural systems from diverse biotic and abiotic stressors (Latham, 2013; Tronsmo et al., 2020). The introduction of synthetic herbicides in the twentieth century revolutionized weed control and substantially increased crop productivity (Powles & Yu, 2010). Owing to their efficiency, selectivity and economic advantages over mechanical or manual weed control, herbicides remain indispensable for global food production (Schulte & Theuvsen, 2015). However, continuous herbicide use has imposed strong selection pressure on weed populations, driving the evolution of resistance. Since the first confirmed case of herbicide resistance in 1957, the number of resistant weed species has increased dramatically and poses a major threat to sustainable agriculture. Currently, resistance to 21 of 31 known herbicide modes of action (MoA) has been documented in 273 species across 75 countries (Heap, 2025). Herbicide resistance arises through genetic variation that enables individual plants to survive and reproduce under herbicide selection pressure. Resistance mechanisms are broadly categorized into non-target-site resistance (NTSR), such as enhanced metabolism or altered translocation, and target-site resistance (TSR), which results from modifications of the herbicide target protein (Gaines et al., 2020). TSR is often conferred by single-nucleotide polymorphisms (SNPs) in the herbicide target gene that reduce binding affinity while maintaining enzymatic function (Gaines et al., 2020; Beckie, 2020; Murphy & Tranel, 2019). Such SNPs can confer resistance to a single herbicide or an entire chemical class, depending on the structural similarity of the compounds and their target interactions. Therefore, early biochemical and structural evaluation of herbicide targets is essential for assessing resistance risk and for guiding the development of sustainable modes of action in weed management.

Grass weeds such as *blackgrass* [*Alopecurus myosuroides* Huds.] and *Italian ryegrass* [*Lolium multiflorum* Lam.] are among the most problematic species in cereal production due to their rapid adaptation and widespread resistance to acetyl-CoA carboxylase (ACCase) and very-long-chain fatty-acid (VLCFA) synthesis inhibitors and ALS herbicides (Messelhäuser et al., 2021; Scarabel et al., 2020). This problem has increased the demand for herbicides with novel biochemical targets and unique MoAs.

Cinmethylin [(1S,2R,4R)-1-methyl-2-[(2-methylphenyl)methoxy]-4-propan-2-yl-7-oxabicyclo[2.2.1]heptane)] represents such an innovation. Discovered by Shell in the 1980s, cinmethylin inhibits fatty acid thioesterases (FATs), key enzymes responsible for terminating plastidial fatty-acid biosynthesis (Campe et al., 2018; Jones et al., 1995). FATs catalyze the hydrolysis of acyl-ACP intermediates and thereby releasing free fatty acids which act as essential precursors for membrane lipids, cutin and suberin (Haslam & Kunst, 2013; Ohlrogge & Browse, 1995). Inhibition of FATs disrupts fatty acid release, leading to impaired lipid metabolism, loss of membrane integrity and ultimately plant death (Campe et al., 2018).

Plant FATs are classified into two families, FAT A and FAT B, which differ in substrate specificity and physiological function and occur in multiple isoforms. FAT A preferentially hydrolyzes unsaturated acyl-ACP substrates, whereas FAT B primarily targets saturated acyl-ACP species (Liu et al., 2022).

Despite its early discovery, cinmethylin was not widely adopted in agricultural practice for several decades due to the dominance of competing herbicide classes targeting ACCase and VLCFA biosynthesis (Campe et al. 2018, Yu et al., 2024). However, the extensive use of these alternative MoAs has resulted in widespread resistance, particularly in problematic grass weeds

such as *A. myosuroides*. As resistance to established herbicides increased, the agronomic relevance of cinmethylin was rediscovered (Campe et al., 2018). For many years, the precise MoA of cinmethylin remained unclear and it was therefore classified by the Herbicide Resistance Action Committee (HRAC) as having an unknown MoA. Recent studies employing cellular target profiling identified FAT isoforms as direct molecular targets of cinmethylin, a finding subsequently confirmed by co-crystallization and biochemical assays using recombinant FAT proteins (Campe et al., 2018). These insights led to the reclassification of cinmethylin as the first herbicide in HRAC Group 30, representing a truly novel MoA. However, the resistance risk associated with this novel target, specifically whether amino acid substitutions in FATs can reduce cinmethylin sensitivity while preserving enzymatic activity, remains insufficiently characterized. Given the ecological and economic impact of herbicide-resistant weeds it is a key challenge to identify target-site mutations that may confer herbicide resistance. *A. myosuroides* and *L. multiflorum* are among the most problematic grass weeds in Europe (Messelhäuser et al., 2021; Scarabel et al., 2020). The objective of this study was to evaluate the resistance potential of cinmethylin by determining whether defined amino acid substitutions in FATs reduce cinmethylin inhibition while maintaining enzymatic activity. To also assess potential cross-species relevance, FAT sequences from both *A. myosuroides* and *L. multiflorum* were compared to identify conserved regions and candidate positions for amino acid substitutions.

Materials and Methods

Study Design

The study was conducted as an exploratory biochemical screening of FAT A and B mutant variants using the *A. myosuroides* backbone sequence. Due to the large number of variants, each mutant was produced once and evaluated under standardized assay conditions alongside wild type controls.

Expression of recombinant FAT proteins

Putative target-site mutations in FATA and FATB were identified by performing molecular docking simulations in which cinmethylin was computationally positioned within the active binding pockets of both enzymes, allowing the prediction of amino acid residues likely to interact with the herbicide and potentially contribute to altered binding.

WT proteins were expressed in five independent biological replicates; each was assayed in duplicate wells (technical replicate). Mutants were expressed once (one biological replicate) and assayed in duplicate wells.

Recombinant proteins were heterologously expressed in *Escherichia coli* (*E. coli*). FAT encoding plasmids (pET24d, c-terminal His-tag) were obtained as synthetic constructs (GeneArt, Thermo Fisher Scientific). Plasmids were transformed into *E. coli* BL21-CodonPlus(DE3)-RIL competent cells (Agilent). Transformants were selected on LB agar containing kanamycin (50 µg mL⁻¹) and chloramphenicol (34 µg mL⁻¹). Single colonies were used to inoculate LB pre-cultures (20 mL) and incubated at 37°C 140 rpm for 17 h. Main cultures were established in autoinduction medium with ZY-base supplemented with trace elements and carbon sources as described in Tab. 1. Cultures were incubated at 37 °C and 140 rpm for 5 h followed by 25 °C at

140 rpm for 21 h. Cells were harvested by centrifugation (7000 x g, 30 min, 4 °C), pellets were frozen on dry ice and stored at -80 °C until purification.

Protein purification (IMAC) and dialysis

Cell pellets were resuspended in lysis buffer (50 mM KH₂PO₄, 150 mM NaCl, pH 7.2) supplemented with BugBuster (10% [v/v]; MilliporeSigma), lysozyme (133 µg mL⁻¹), DNase I (1 mg mL⁻¹) and EDTA-free protease inhibitor (cOmplete™, Roche). Lysis was performed for 20 min at room temperature (at 180 rpm). Soluble fractions were purified using Ni-NTA agarose resin (HisPur™ Superflow, Thermo Fisher Scientific) in centrifuge columns (Pierce™, Thermo Fisher Scientific). Columns were equilibrated with lysis buffer, washed with washing buffer (50 mM KH₂PO₄, 150 mM NaCl, 20 mM Imidazole, pH 7.2) and proteins eluted with elution buffer (50 mM KH₂PO₄, 150 mM NaCl, 200 mM Imidazole, pH 7.2). To remove imidazole, eluates were dialyzed overnight at 4 °C against dialysis buffer (5 mM KH₂PO₄, 15 mM NaCl, pH 7.2). Protein concentrations were determined via spectrophotometer (ScanDrop², Analytic Jena). 8-10% (v/v) Glycerol was added prior to storage and samples were stored at -80 °C.

Fluorescence-based acyl-CoA assay and cinmethylin dose-response

FAT activity was quantified using a fluorescence assay based on release of CoA-SH from acyl-CoA substrates, followed by reaction with the thiol-reactive fluorophore CPM (7-diethylamino-3-(4-maleimidophenyl)-4-methylcoumarin). Reactions were conducted in 96-well plates and measured as endpoint fluorescence (excitation 390 nm, emission 470 nm) using a plate reader (CLARIOstar3 plus, BMG Labtech).

Reactions were conducted in assay buffer (50 mM Tris, 500 mM NaCl, 5 mM EDTA, pH 7.2) Oleoyl-CoA was used as substrate for FAT A and palmitoyl-CoA for FAT B at a final concentration of 100 μ M. Cinmethylin stock solutions were prepared in DMSO and serially diluted (1:4) to generate final test concentrations ranging from 1×10^{-4} M to 3.81×10^{-10} M. Reactions were assembled by combining enzyme, compound (or negative control), substrate and CPM detection reagent with standardized incubation times (compound pre-incubation [600 s], CPM incubation [300 s]). Assay assembly was performed using an automated workstation (Biomek NX^P, Beckman Coulter) at 10 °C for enzyme and substrate plates. The final enzyme concentration in the assay was 800 nM. Each variant was measured in two technical replicate wells.

SDS-PAGE verification

Selected purified proteins were analyzed by SDS-PAGE using 10% gels (Mini-PROTEAN TGX Stain-FreeTM, Bio-Rad). Samples were denaturated (95 °C for 10 min) in LDS sample buffer (NuPAGETM, Invitrogen) and run at 139 V for 60-90 min in Mini-PROTEAN Tetra system (Bio-Rad). Gels were imaged with ChemiDocTM MP system (Bio-Rad).

Data processing and statistical analysis

Raw fluorescence values were background corrected by subtracting the negative control signal. Inhibition at the highest cinmethylin concentration (10^{-4} M) was calculated as:

$$\text{Inhibition (\%)}: 100 - \left(\frac{100 \cdot F_{\text{net, with compound}}}{F_{\text{net, no compound}}} \right)$$

Residual activity (%) was determined relative to WT controls within the same batch:

$$\text{Residual activity (\%)}: \frac{F_{\text{net, mutant}}}{F_{\text{net, wild type}}} \cdot 100$$

179 Variants exhibiting <50% residual activity relative to WT were classified as low activity variants
180 and excluded from further sensitivity classification, as such activity levels are assumed to be
181 unlikely to be compatible with plant viability.

182 Dose-response curves and IC₅₀ values were estimated by nonlinear regression using
183 GraphPadPrism (version 10.6.1). IC₅₀ values function as comparative screening.

184 Resistant index was calculated as:

$$\text{Resistance-Index: } \frac{IC_{50, \text{mutant}}}{IC_{50, \text{WT}}}$$

186 Mutants with IC₅₀ values close to WT or below were classified as potentially sensitive to
187 cinmethylin, whereas mutants with increased IC₅₀ relative to WT were classified as potentially
188 resistant.

190 **Cross-species sequence analysis (*A. myosuroides* vs. *L. multiflorum*)**

191 Coding sequences of FAT gene from *A. myosuroides* and *L. multiflorum* were aligned to assess
192 sequence conservation and to map potential mutation positions across species. Alignments were
193 performed with Geneious Prime (Biomatters Ltd.).

194 The mutation probability calculation provides a theoretical estimate of mutation probability
195 based on single nucleotide substitutions and does not account for transition-transversion biases or
196 species specific mutation spectra. Each nucleotide was assumed to mutate with the equal
197 probability into one of the three alternative bases, resulting in a assumed per-base mutation
198 probability of 1/3. At the codon level, three nucleotide positions are available, each with three
199 possible single-nucleotide substitutions which yields a total of nine possible point mutations per
200 codon. If a single point mutation leads to the target codon, the probability was calculated as 1/9;

if two distinct point mutations result in the target codon, the probability was calculated as 2/9 and so on.

Results and Discussion

FAT A & B WT assay and reference sensitivity

Assay performance and reference sensitivity to cinmethylin were established by using wild type (WT) FAT A and B proteins (n = 5 biological replicates each). FAT A WT showed a mean activity of 31218 ± 2813.3 relative fluorescence units (RFU) and a mean inhibition of $90.5\% \pm 2.19$ at saturating cinmethylin concentration (10^{-4}M) (Fig. 1 and Tab. 3). SDS-PAGE confirmed the expected FAT A band at approximately 39 kDa (Fig. 2). Dose-response curves were generated, and IC_{50} -values were estimated with a mean of $3.02 \times 10^{-8} \text{ M} \pm 3.88 \times 10^{-9}$ (Fig. 3; Tab. 2).

FAT B WT showed a mean activity of 129284.5 ± 9373.3 RFU and a mean inhibition of $91.5\% \pm 6.95$ at cinmethylin saturation (Fig. 4 and Tab. 3). SDS-PAGE confirmed the expected FAT B band at approximately 47 kDa (Fig. 5). The mean FAT B WT IC_{50} was and of $1.64 \times 10^{-7} \text{ M} \pm 0.84 \times 10^{-7}$ (Fig. 6; Tab. 2). Together, these results confirm robust assay performance and provide baseline sensitivity values for subsequent mutant screening.

FAT A mutant screening and identification of putative reduced sensitivity candidates

A total of 70 FAT A mutant variants were screened via the acyl-CoA assay. Relative activity in the absence and presence of cinmethylin (10^{-4}M) is shown in (Fig. 7). Residual activity was calculated relative to WT reference. Variants with residual activity $<50\%$ of WT were excluded from further sensitivity determination. Among the remaining variants the three mutants R171Q,

R171I and R171M showed low inhibition at saturating cinmethylin concentration (24.3%, 15.5%, 11.00% inhibition) while maintaining residual activity >50% of WT (Tab. 3). This indicates a significant reduced sensitivity at saturating inhibitor concentration and highlights residue R171 as a critical position for cinmethylin-mediated inhibition. All other variants exceeding the residual activity threshold showed inhibition greater than 50% at 10^{-4} M and were therefore advanced to dose-response analysis for IC₅₀ estimation.

FAT A dose-response analysis and magnitude of sensitivity shifts

Dose-response curves and IC₅₀ estimates for prioritized FAT A mutant variants are shown in Fig. 8 and IC₅₀ values were compared against WT baseline and summarized as resistance indices (Tab. 3). The largest shift relative to WT was observed for R171K (IC₅₀ 1.56×10^{-5} M; resistance index 516.6). Additional shifts were observed for W173L (2.05×10^{-7} M; resistance index 6.77), H112Q (1.28×10^{-7} M; resistance index 4.24) and F118C (8.87×10^{-8} M; resistance index 2.94). Approximately twofold increases in IC₅₀ were observed for T127K (1.99-fold) and F118Y (2.04-fold). Other variants displayed IC₅₀ values comparable to WT. R171Q, R171I and R171M showed the strongest insensitivity levels with no measurable IC₅₀. These results are consistent with cinmethylin binding to FAT enzymes and suggest that specific substitutions can reduce inhibitor potency, in line with prior structural and biochemical evidence that cinmethylin targets acyl-ACP thioesterases (Campe et al., 2018).

Codon-based mutation probability assessment for prioritized FAT A variants in A. myosuroides

To contextualize the evolutionary plausibility of the observed sensitivity shifts, codon-level nucleotide polymorphism (NP) requirements were determined for nine prioritized FAT A

variants (H112Q, F118Y, F118C, T127K, R171K, R171Q, R171I, R171M, W173L) (Tab. 4). Variants required one, two or three NPs depending on the codon changes. In the defined and simplified model (equal probability of the three alternative bases per position), mutations requiring a single-NP (SNP) represent the most likely class at the population level in this framework, whereas two- and three-NP changes were orders of magnitude less likely to occur. For SNP candidates, codon substitution mappings were generated (Tab. 5). For H112Q, two SNP-derived target codons (CAA or CAG) are possible, leading to an increased within-codon probability (2/9) of reaching a glutamine codon under the model assumptions. While this model does not account for any transition-transversion bias, selection on protein stability etc., it provides a first order ranking of mutational accessibility and supports prioritization of candidates for further monitoring.

Cross-species conservation of FAT A positions in Lolium multiflorum

Because *A. myosuroides* and *L. multiflorum* are agronomically important amongst other things due to their rapid resistance evolution, conservation of prioritized FAT A positions was assessed. FAT A sequences showed high similarity (91.1%). Alignment of the regions of the prioritized positions (H112, R171, W173, T127, F118) indicated conservation of the corresponding amino acids in *L. multiflorum* (Tab. 6). At the cDNA level, codons were identical between both species for H112, R171, W173 and T127, while F118 differed. Consequently, codon-specific NP requirements were recalculated for *L. multiflorum* for F118Y and F118C and SNP substitution mapping confirmed that model assumptions remained applicable for these targets (

Tab. 7). These findings suggest that mutational routes identified in *A. myosuroides* may also be accessible in *L. multiflorum*.

FAT B mutant screening and dose response analysis

A total of 34 FAT B mutant variants were screened via the acyl-CoA assay (Fig. 9). Variants with residual activity <50% of WT were excluded from further sensitivity classification. The remaining mutants and their inhibition values are summarized in Tab. 3. All listed variants exceeded 50% inhibition at 10^{-4} M were advanced to dose-response analysis (Fig. 10). IC_{50} values are shown in Tab. 3 as well as resistance indices relative to WT. The largest shift was observed for P192R (IC_{50} 1.60×10^{-6} M; resistance index: 9.76). V178G (IC_{50} 6.24×10^{-7} M; resistance index: 3.80) and A190E (IC_{50} 4.15×10^{-7} M; resistance index: 2.53) also showed increased IC_{50} values relative to WT. A181V displayed an approximately twofold increase (1.93-fold). In contrast, V226C and the double mutant A181V&V226C exhibited lower IC_{50} estimates than WT, which indicates that not all substitutions reduce cinmethylin sensitivity and that certain changes may increase apparent susceptibility under the assay conditions.

Codon-based mutation probability assessment for prioritized FAT B variants and cross-species context

For the prioritized FAT B variants (A181V, V178G, A190E, P192R), required NPs to reach target codons in *A. myosuroides* were determined (Tab. 8). SNP substitution profiles were

generated for selected variants (Tab. 9) yielding a target-codon probability of 1/9 (11.11%) for the analyzed variants under model assumptions.

A. myosuroides and *L. multiflorum* were highly similar for FAT B (93.2%). Alignment indicated conservation of the critical amino acids (A181, V178, A190, P192) across species (Tab. 10) Codon comparisons revealed differences for A181 and P192 which leads to recalculation of NP requirements (

Tab. 7).

Implications for cinmethylin durability and resistance risk

Herbicide resistance is a major challenge in modern agriculture, particularly in grass weeds with large population sizes and strong selection pressure (Kersten et al., 2023). Cinmethylin represents an important alternative mode of action through inhibition of fatty acid thioesterases, key enzymes in the de novo fatty acid biosynthesis that are essential for plant viability (Campe et al., 2018; Ohlrogge & Browse, 1995).

The findings of this study provide first indications that cinmethylin, despite the absence of reported resistance cases to date, is not immune to potential resistance evolution. Specific amino acid substitutions in the target proteins FAT A and FAT B can reduce cinmethylin sensitivity in vitro, which indicates that target-site resistance is theoretically possible. However, the magnitude of the observed IC₅₀ shifts was generally moderate.

In weed species with high population sizes, such as *Alopecurus myosuroides* and *Lolium multiflorum*, the emergence of resistant individuals under repeated cinmethylin selection can not be excluded. The high degree of conservation observed at critical amino acid positions between these species further suggests that resistance-conferring substitutions arising in one species may analogously evolve in another.

At the molecular level, the present screening indicates that specific target-site substitutions in FAT A and B can reduce cinmethylin sensitivity in vitro, with the strongest effects observed at FAT A residue R171 and the FAT B substitution P192R. Structural evidence supports these observations, as cinmethylin binding involves interactions within a defined pocket in which a conserved arginine residue contributes to hydrogen bonding and ligand stabilization (Campe et al., 2018). Substitutions at R171 alter the side-chain geometry and therefore weaken the binding affinity to the compound.

Importantly, among the many mutant variants tested, only a small number retained at least 50% residual enzymatic activity and would therefore be expected to remain viable in plants. While substitutions at R171 exhibited the strongest reduction in cinmethylin binding and therefore represent the greatest theoretical resistance risk, they are extremely unlikely to occur under field conditions, as they would require multiple specific nucleotide polymorphisms. By contrast, the H112Q substitution shows a more realistic change, its effect on cinmethylin binding was comparatively modest.

Overall, this study demonstrates that while cinmethylin represents an important herbicide with a novel mode of action and no reported resistance cases to date, its efficacy could in theory be affected by specific target-site mutations in FAT enzymes, particularly those arising from SNPs.

However, the findings clearly demonstrate that while theoretical resistance routes exist, the practical risk remains low, underscoring cinmethylin robustness as a reliable herbicide option. The high sequence conservation between *Alopecurus myosuroides* and *Lolium multiflorum* underscores the agronomic relevance of these findings, as resistance evolution in both species appears possible under selection pressure. This highlights the necessity for proactive, integrated resistance management to safeguard its long-term effectiveness.

Limitations

Although the results of the study provide important insights, the mutations analyzed were tested only in vitro and their actual impact on resistance in plants remains to be determined. While these provide theoretical insights, they cannot capture the complex interplay of metabolism, fitness costs and selection pressure occurring in field populations. Future work should therefore include ectopic expression in model species such as *Arabidopsis thaliana* to assess in vivo functionality.

The codon-based probability model applied in this study represents a strongly simplified framework. It assumes equal mutation probabilities for all base substitutions and does not account for biological realities. As a result, the calculated probabilities must be interpreted as approximations rather than absolute values.

Nevertheless, the combined analysis presented here provides a valuable framework for assessing the potential risk of resistance evolution against cinmethylin.

Acknowledgments

The author thanks Dr. Aimone Porri (BASF Agricultural Center, Limburgerhof, Germany) for supervision and guidance throughout the research project and department of phytopathology at RPTU Kaiserslautern for academic support. Technical assistance from BASF laboratory staff is gratefully acknowledged.

Funding

This research received no specific grant from any funding agency, commercial or non-profit sectors. The work was conducted as part of a bachelor research project in collaboration with BASF Agricultural Center, Limburgerhof, Germany.

Competing Interests

The author conducted this research as part of a bachelor thesis project in collaboration with BASF Agricultural Center, Limburgerhof, Germany. The author declares no other competing interests.

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Tables

Tab. 1 Media formulations used for recombinant expression of *A. myosuroides* FAT proteins in *E.coli*

Component	Final concentration
LB pre-culture medium	
Tryptone	10g L ⁻¹
Yeast extract	5g L ⁻¹
NaCl	10g L ⁻¹
Kanamycin	50µg mL ⁻¹
Chloramphenicol	34µg mL ⁻¹
Autoinduction medium ZY based	
Tryptone	10.8 g L ⁻¹
Yeast extract	5.4 g L ⁻¹
MgSO ₄ (1 M)	1 mM
Trace elements, including:	
FeCl ₃ x 6H ₂ O (0,05 M)	50 µM L ⁻¹
CaCl ₂ x 2H ₂ O (0,02 M)	20 µM L ⁻¹
MnCl ₂ x 4H ₂ O (0,01 M)	10 µM L ⁻¹
ZnSO ₄ x 7H ₂ O (0,01 M)	10 µM L ⁻¹
CoCl ₂ x 6H ₂ O (0,002 M)	2 µM L ⁻¹
CuCl ₂ x 2H ₂ O (0,002 M)	2 µM L ⁻¹
NiCl ₂ x 6H ₂ O (0,002 M)	2 µM L ⁻¹
Na ₂ MoO ₄ x 2H ₂ O (0,002 M)	2 µM L ⁻¹
50x5052, containing	
Glycerol	0.5% (v/v)
Glucose	0.05% (v/v)
Lactose	0.2% (v/v)
50xM, containing:	
NH ₄ Cl (2,5 M)	50 mM
KH ₂ PO ₄ (1,25 M)	25 mM
Na ₂ HPO ₄ x 2H ₂ O (1,25 M)	25 mM
Na ₂ SO ₄	5 mM
Kanamycin	50µg mL ⁻¹
Chloramphenicol	34µg mL ⁻¹
Pre-culture	2% (v/v)

Tab. 2 IC₅₀ values of *Alopecurus myosuroides* FAT A and B wild-type proteins determined from five biological replicates each

Replicate	IC ₅₀ [M]
FAT A WT 1	2,89 x 10 ⁻⁸
FAT A WT 2	2,91 x 10 ⁻⁸
FAT A WT 3	3,64 x 10 ⁻⁸
FAT A WT 4	3,07 x 10 ⁻⁸
FAT A WT 5	2,59 x 10 ⁻⁸
Mean & SD	3,02 x 10 ⁻⁸ ± 3,88 x 10 ⁻⁹
FAT B WT 1	1,17 x 10 ⁻⁷
FAT B WT 2	1,06x 10 ⁻⁷
FAT B WT 3	3,10 x 10 ⁻⁷
FAT B WT 4	1,28 x 10 ⁻⁷
FAT B WT 5	1,58 x 10 ⁻⁷
Mean & SD	1,64 x 10 ⁻⁷ ± 0,84 x 10 ⁻⁷

Tab. 3 *Alopecurus myosuroides* FAT A and B mutants with residual activity >50 % and corresponding inhibition values at cinmethylin saturation 10^{-4} M.

Mutant	Residual activity [%]	Inhibition [%]	IC ₅₀	Resistance-Index
FAT A				
WT	n.d.	90.50	$3,02 \times 10^{-8}$	n.d.
H112Q	185.02	87.87	1.28×10^{-7}	4.24
V116L	50.58	81.87	4.85×10^{-8}	1.61
F118Y	236.41	96.61	6.17×10^{-8}	2.04
F118C	222.73	52.88	8.87×10^{-8}	2.94
A124G	78.32	90.13	3.44×10^{-8}	1.14
A124S	92.77	91.60	3.77×10^{-8}	1.25
T127K	65.73	86.00	6.02×10^{-8}	1.99
R171K	60.87	56.56	1.56×10^{-5}	516.56
R171Q	54.72	24.32	n.d.	n.d.
R171I	84.70	15.53	n.d.	n.d.
R171M	57.39	10.96	n.d.	n.d.
W173L	329.68	91.63	2.05×10^{-7}	6.77
W173G	50.96	79.33	3.79×10^{-8}	1.25
FAT B				
WT	n.d.	91.5	1.64×10^{-7}	n.d.
A181V	66.07	85.96	3.17×10^{-7}	1.93
V226C	125.77	87.61	9.47×10^{-8}	0.58
A181V&V226C	79.39	89.56	1.23×10^{-7}	0.75
V178A	114.36	92.89	1.61×10^{-7}	0.98
V178G	192.44	94.90	6.24×10^{-7}	3.80
A190E	146.85	93.35	4.15×10^{-7}	2.53
P192R	111.02	92.36	1.60×10^{-6}	9.76

Tab. 4 Required nucleotide polymorphisms to reach target codons resulting in respective amino acid exchange in *Alopecurus myosuroides* FAT A

Mutation	Original codon	Target Codon	Required substitution		Required NPs
H112Q	CAT	CAA	T	A	1
		CAG	T	G	1
R171K	CGT	AAA	C	A	3
			G	A	
			T	A	
		AAG	C	A	3
			G	A	
			T	G	
W173L	TGG	CTT	T	C	3
			G	T	
			G	T	
		CTC	T	C	3
			G	T	
			G	C	
		CTA	T	C	3
			G	T	
			G	A	
		CTG	T	C	2
			G	T	
		TTG	G	T	1
T127K	ACA	AAG	G	T	2
			G	A	
		AAA	C	A	1
			C	A	
F118Y	TTC	TAT	T	A	2
			C	T	
F118C	TTC	TAC	T	A	1
		TGT	T	G	2
R171Q	CGT	TGC	C	T	
			G	T	1
		CAA	G	A	2
			T	A	
R171I	CGT	CAG	G	A	2
			T	G	
		ATT	C	A	2
			G	T	
		ATC	C	A	3
			G	T	
R171M	CGT	ATA	T	C	3
			G	T	
		ATG	C	A	3
			G	T	

Tab. 5 cDNA-based analysis of *Alopecurus myosuroides* FAT A, showing possible nucleotide substitutions, resulting codons and amino acids and the probability of occurrence of the target mutation

Mutation	Base	Possible Substitutionen	Resulting Codon	Resulting Aminoacid	Probability [%]
H112Q	C	A	AAT	N	22.22
		G	GAT	D	
		T	TAT	Y	
	A	C	CCT	P	
		G	CGT	R	
		T	CTT	L	
	T	A	CAA	Q	
		C	CAC	H	
		G	CAG	Q	
F118Y	T	A	ATC	I	11.11
		C	CTC	L	
		G	GTC	V	
	T	A	TAC	Y	
		C	TCC	S	
		G	TGC	C	
	C	A	TTA	L	
		G	TTG	L	
		T	TTT	F	
F118C	T	A	ATC	I	11.11
		C	CTC	L	
		G	GTC	V	
	T	A	TAC	Y	
		C	TCC	S	
		G	TGC	C	
	C	A	TTA	L	
		G	TTG	L	
		T	TTT	F	
A181V	G	A	ACC	T	11.11
		C	CCC	P	
		T	TCC	S	
	C	A	GAC	D	
		G	GGC	G	
		T	GTC	V	
	C	A	GCA	A	
		G	GCG	A	
		T	GCT	A	
V178G	G	A	ATG	M	11.11
		C	CTG	L	
		T	TTG	L	
	T	A	GAG	E	
		G	GGG	G	
		C	GCG	A	
	G	A	GTA	V	
		C	GTC	V	
		T	GTT	V	
P192R	C	A	ACC	T	11.11
		G	GCC	A	
		T	TCC	S	
	C	A	CAC	H	
		G	CGC	R	
		T	CTC	L	
	C	A	CCA	P	
		G	CCG	P	
		T	CCT	P	

Tab. 6 Alignments showing the consensus identity of relevant protein regions in *Alopecurus myosuroides* and *Lolium multiflorum* FAT A

Region	Alignment
H112	Consensus Identity 1. ALOMY F... 2. Lolium F... 112 aa & 2 gaps^x
R171	Consensus Identity 1. ALOMY F... 2. Lolium F... 171 aa & 2 gaps^x
W173	Consensus Identity 1. ALOMY F... 2. Lolium F... 173 aa & 2 gaps^x
T127	Consensus Identity 1. ALOMY F... 2. Lolium F... 127 aa & 2 gaps^x
F118	Consensus Identity 1. ALOMY F... 2. Lolium F... 118 aa & 2 gaps^x

Tab. 7 cDNA-based analysis of *Lolium multiflorum* FAT A and B, showing possible nucleotide substitutions, resulting codons and amino acids and the probability of occurrence of the target mutation

Mutation	Base	Possible Substitutionen	Resulting Codon	Resulting Aminoacid	Probability [%]
FAT A					
F118Y	T	A	ATT	I	11,11
		C	CTT	L	
		G	GTT	V	
	T	A	TAT	Y	
		C	TCT	S	
		G	TGT	C	
	T	A	TTA	L	
		C	TTC	F	
		G	TTG	L	
F118Y	T	A	ATT	I	11,11
		C	CTT	L	
		G	GTT	V	
	T	A	TAT	Y	
		C	TCT	S	
		G	TGT	C	
	T	A	TTA	L	
		C	TTC	F	
		G	TTG	L	
FAT B					
A181V	G	A	ACT	T	11,11
		C	CCT	P	
		T	TCT	S	
	C	A	GAT	D	
		G	GGT	G	
		T	GTT	V	
	T	A	GCA	A	
		G	GCG	A	
		C	GCC	A	
P192R	C	A	ACG	T	11,11
		G	GCG	A	
		T	TCG	S	
	C	A	CAG	Q	
		G	CGG	R	
		T	CTG	L	
	G	A	CCA	P	
		C	CCC	P	
		T	CCT	P	

Tab. 8 Required nucleotide polymorphisms to reach target codons resulting in respective amino acid exchange in *Alopecurus myosuroides* FAT B

Mutation	Original Codon	Target Codon	Required substitution (Before / After)		Required NPs
A181V	GCC	GTG	C	T	2
			C	G	
		GTA	C	T	2
			C	A	
		GTC	C	T	1
		GTT	C	T	2
			C	T	
V178G	GTG	GGG	T	G	1
		GGA	T	G	2
			G	A	
		GGC	T	G	2
			G	C	
		GGT	T	G	2
			G	T	
A190E	GCT	GAG	C	A	2
			T	G	
		GAA	C	A	2
			T	A	
P192R	CCC	CGG	C	G	2
			C	G	
		CGA	C	G	2
			C	A	
		CGC	C	G	1
		CGT	C	G	2
			C	T	
		AGG	C	A	3
			C	G	
			C	G	
		AGA	C	A	3
			C	G	
			C	A	

Tab. 9 cDNA-based analysis of *Alopecurus myosuroides* FAT A, showing possible nucleotide substitutions, resulting codons and amino acids and the probability of occurrence of the target mutation

Mutation	Base	Possible Substitutionen	Resulting Codon	Resulting Aminoacid	Probability [%]
A181V	G	A	ACC	T	11,11
		C	CCC	P	
		T	TCC	S	
	C	A	GAC	D	
		G	GGC	G	
		T	GTC	V	
	C	A	GCA	A	
		G	GCG	A	
		T	GCT	A	
V178G	G	A	ATG	M	11,11
		C	CTG	L	
		T	TTG	L	
	T	A	GAG	E	
		G	GGG	G	
		C	GCG	A	
	G	A	GTA	V	
		C	GTC	V	
		T	GTT	V	
P192R	C	A	ACC	T	11,11
		G	GCC	A	
		T	TCC	S	
	C	A	CAC	H	
		G	CGC	R	
		T	CTC	L	
	C	A	CCA	P	
		G	CCG	P	
		T	CCT	P	

Tab. 10 Alignments showing the consensus identity of relevant protein regions in *Alopecurus myosuroides* and *Lolium multiflorum* FAT B

Region	Alignment
A181	Consensus Identity 170180182 et Asn His Met Gln Glu Thr Ala Leu Asn His Val Lys Thr Ala 1. ALOMY F... 2. Lolium F... 181 aa & 1 gap^x
V178	Consensus Identity 170179 Thr Leu Met Asn His Met Gln Glu Thr Ala Leu Asn His Val 1. ALOMY F... 2. Lolium F... 178 aa & 1 gap^x
A190	Consensus Identity 180191 Leu Asn His Val Lys Thr Ala Gly Leu Leu Gly Asp Gly Phe Gly Ala T 1. ALOMY F... 2. Lolium F... 190 aa & 1 gap^x
P192	Consensus Identity 180190193 s Val Lys Thr Ala Gly Leu Leu Gly Asp Gly Phe Gly Ala Thr Pro 1. ALOMY F... 2. Lolium F... 192 aa & 1 gap^x

Figures

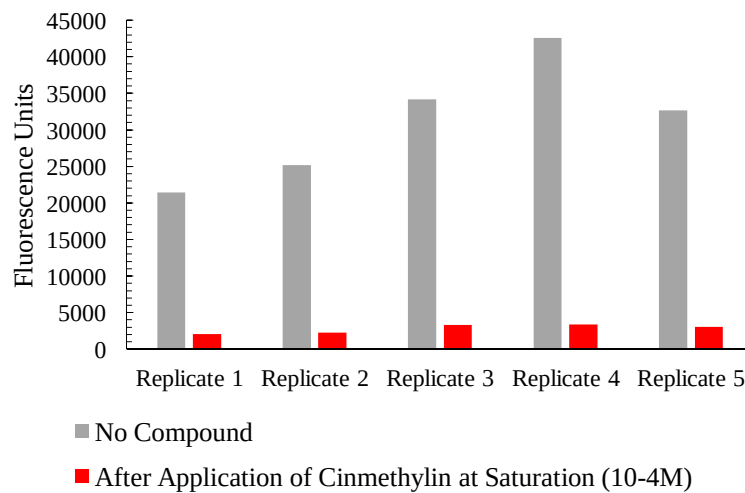


Fig. 1 Acyl-CoA Assay of FAT A from *Alopecurus myosuroides* WT biological replicates with and without cinmethylin treatment. Grey bars represent relative in the absence of compound, while red bars show inhibited after application of cinmethylin (10^{-4} M).

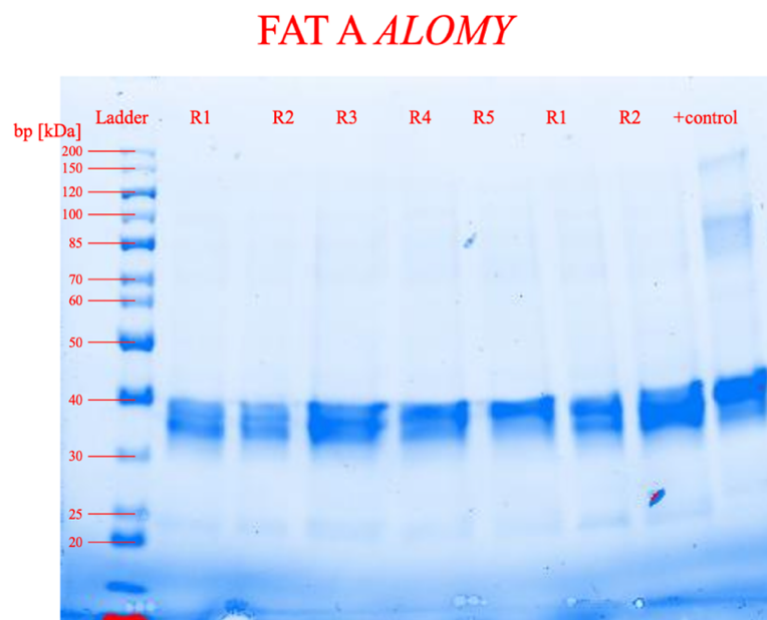


Fig. 2 SDS-PAGE of *Alopecurus myosuroides* FAT A with five replicates (including two loaded in duplicate). As a positive control an independently and elaborately purified FAT A protein.

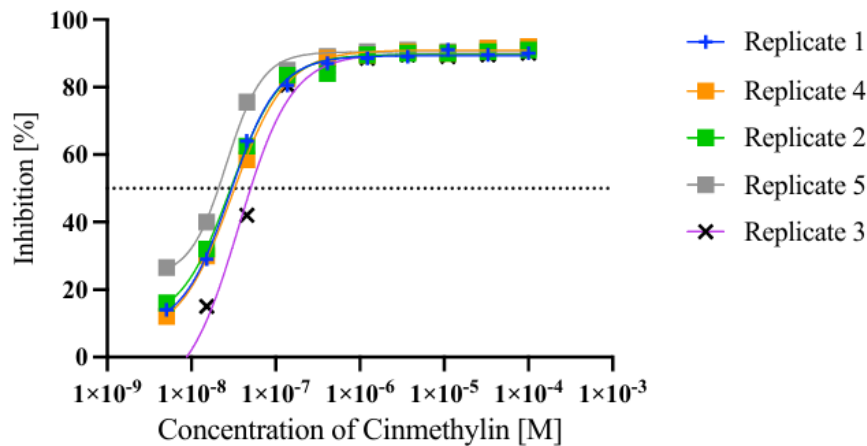


Fig. 3 Determination of IC₅₀ values for *Alopecurus myosuroides* FAT A WT in response to cinmethylin

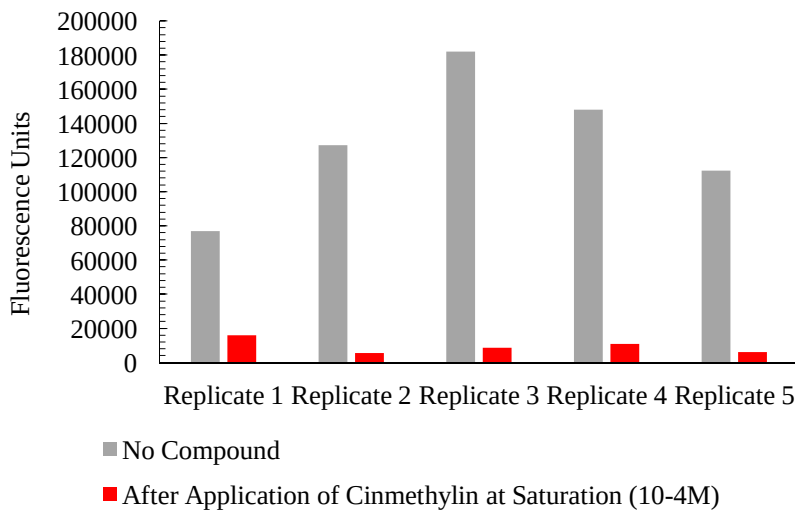


Fig. 4 Acyl-CoA Assay of FAT B from *Alopecurus myosuroides* WT biological replicates with and without cinmethylin treatment. Grey bars represent relative in the absence of compound, while red bars show inhibited after application of cinmethylin (10^{-4} M).

FAT B *ALOMY*

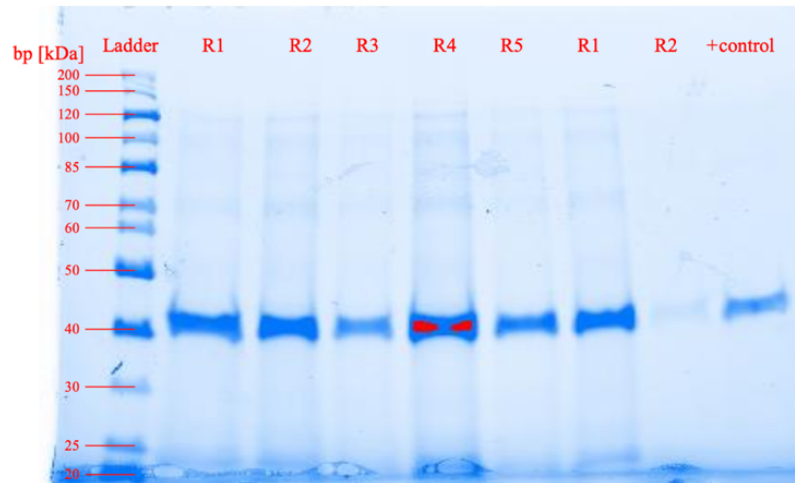


Fig. 5 **SDS-PAGE of *Alopecurus myosuroides* FAT B** with five replicates (including two loaded in duplicate). As positive control an independently and elaborately purified FAT A protein (provided by Marnet, BASF) was included.

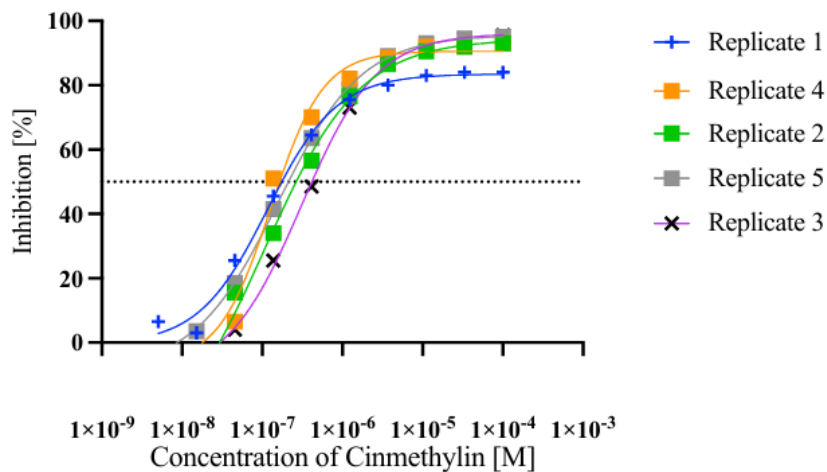
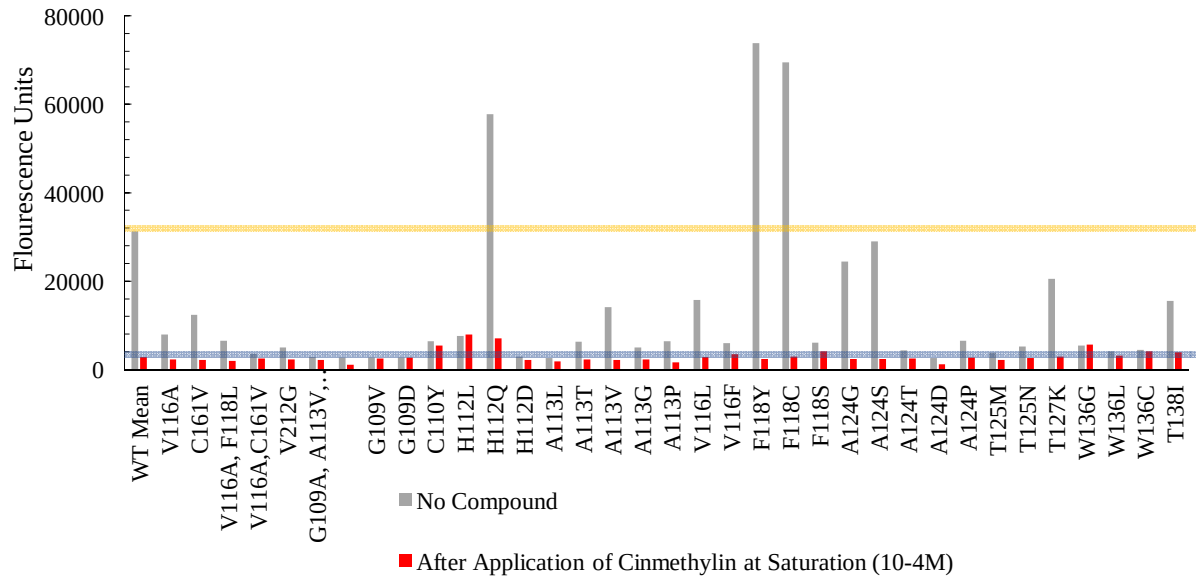


Fig. 6 **Determination of IC₅₀ values for *Alopecurus myosuroides* FAT B WT in response to cinmethylin**

A



B

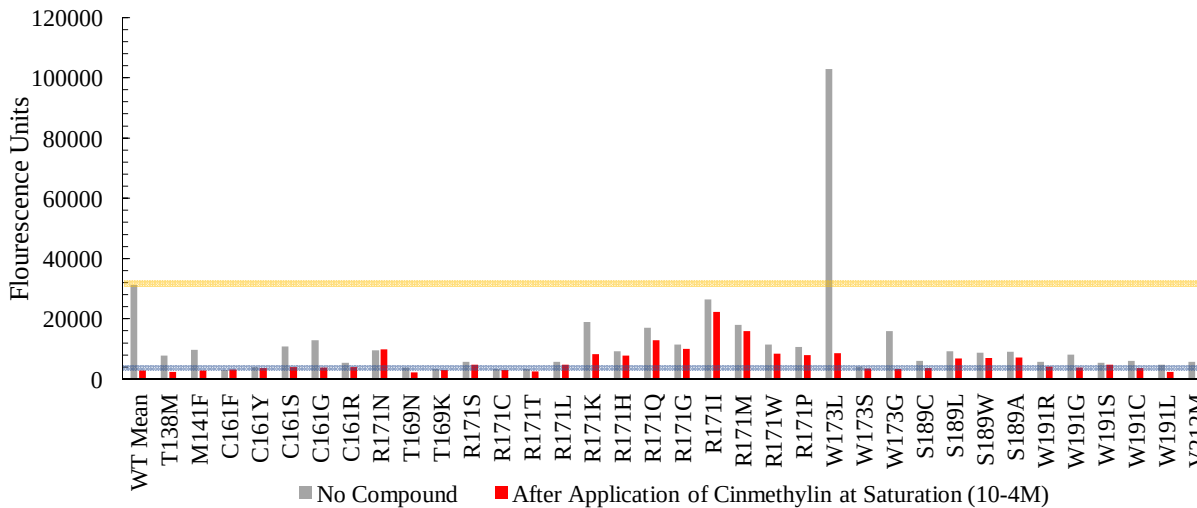
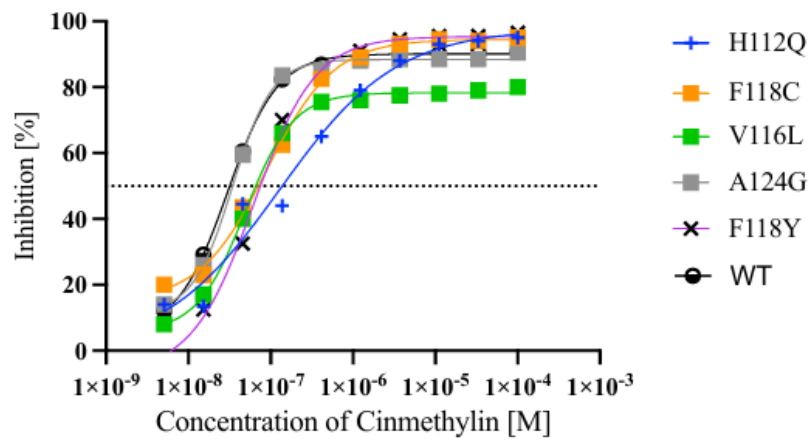


Fig. 7 **Acyl-CoA Assay of FAT A mutant of *Alopeurus myosuroides* variants before and after cinmethylin treatment.** Grey bars represent relative activity without compound application, while red bars show residual activity after application of cinmethylin (10^{-4} M). The horizontal lines indicate the reference level corresponding to the WT.

A



B

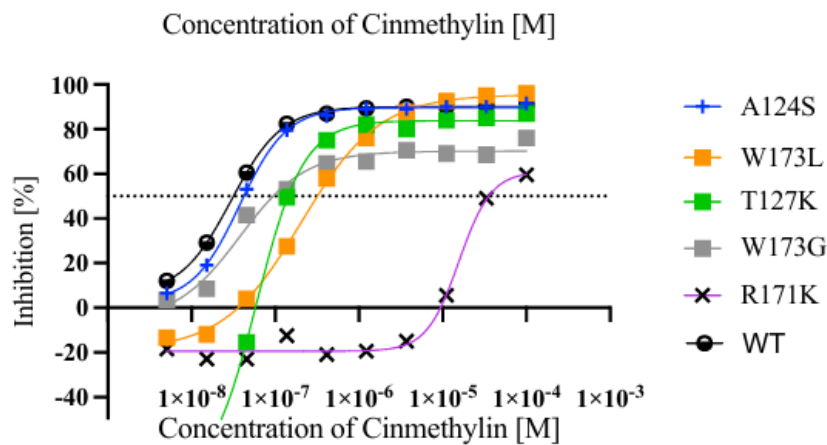


Fig. 8 Dose-response curves of *Alopecurus myosuroides* FAT A mutants for (A) H112Q, V116L, F118Y, F118C, A124G and (B) A124S, T127K, W173G, R171K.

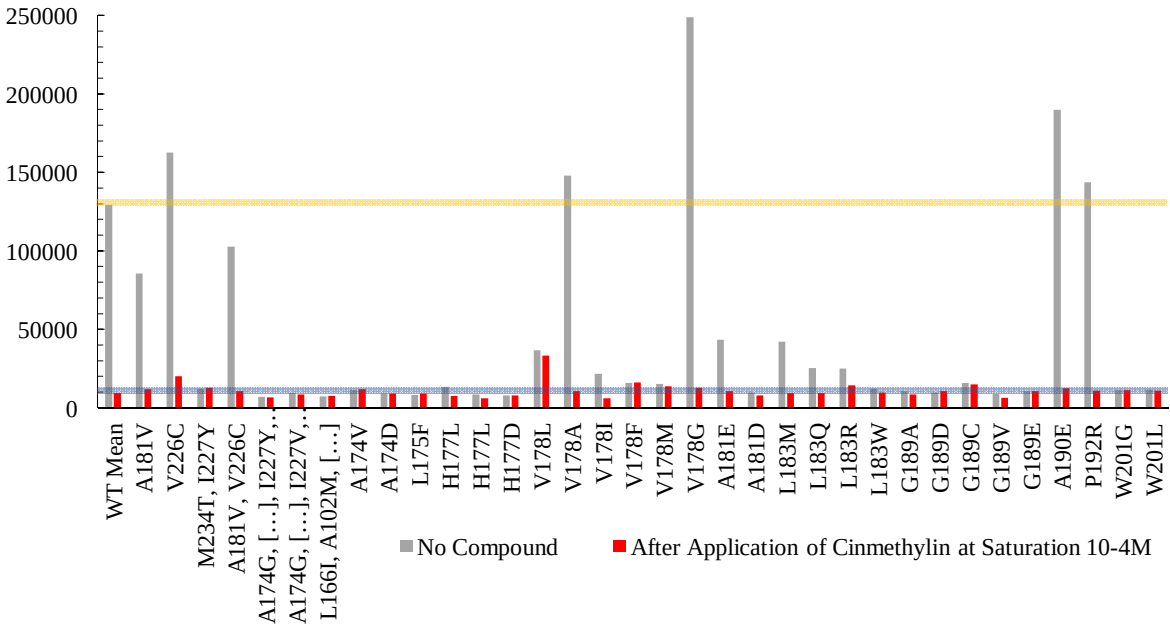


Fig. 9 **Acyl-CoA Assay of *Alopecurus myosuroides* FAT B mutant variants before and after cinmethylin application.** Grey bars represent relative activity without compound application, while red bars show activity after application of cinmethylin (10^{-4} M). The horizontal lines indicate the reference level corresponding to the WT.

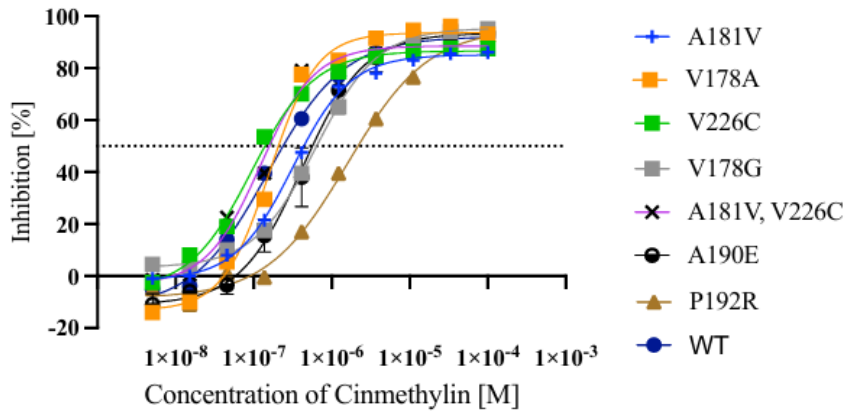


Fig. 10 **Dose-response curves of *Alopecurus myosuroides* FAT B mutants.**

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Mutant	Protein concentration [mg/ml]	FU (- compound)	FU (+compound)	Residual activity [%]	Inhibition [%]
FAT A Generation 1					
WT	6,47	31218	2813,3	n.d.	90,5
V116A	5,68	7954	2238	25,48	71,86
C161V	7,17	12327	2187	39,49	82,26
V116A, F118L	6,89	6544	1996	20,96	69,50
V116A,C161V	5,19	3576	2466	11,45	31,04
V212G	6,35	4945	2269	15,84	54,12
G109A, A113V, V116A, F118L, A124G, C161V, V212I	8,56	2817	2168	9,02	23,04
I101L, G109A, C110L, A113V, V116A, F118L, S119L, T120G, A124G, T125A, T127P, T138V, C161V, M167A, T169M, S205P, V212I, I214P	7,73	2720	1097	8,71	59,67
G109V	10,45	2746	2485	8,80	9,50
G109D	6,69	2713	2662	8,69	1,88
C110Y	7,12	6388	5438	20,46	14,87
H112L	0,89	7600	7901	24,34	-3,96
H112Q	7,96	57760	7007	185,02	87,87
H112D	6,4	2949	2202	9,45	25,33
A113L	5,02	2600	1838	8,33	29,31
A113T	7,9	6301	2223	20,18	64,72
A113V	5,97	14133	2168	45,27	84,66
A113G	5,45	4971	2327	15,92	53,19
A113P	5,51	6382	1668	20,44	73,86
V116L	6,19	15789	2862	50,58	81,87
V116F	7,7	5971	3507	19,13	41,27
F118Y	5,66	73802	2357	236,41	96,61
F118C	2,42	69532	2881	222,73	52,88
F118S	6,86	6114	4142	19,58	32,25
A124G	5,55	24450	2413	78,32	90,13
A124S	6,86	28962	2433	92,77	91,60
A124T	6,08	4355	2505	13,95	42,48
A124D	6,26	2559	1179	8,20	53,93
A124P	3,22	6491	2672	20,79	58,84
T125M	3,97	3798	2219	12,17	41,57
T125N	4,72	5163	2632	16,54	49,02
T127K	2,87	20519	2873	65,73	86,00
W136G	7,42	5433	5607	17,40	-3,20
W136L	0,22	4114	3114	13,80	24,31
W136C	4,23	4433	4095	14,20	7,62

T138I	3,04	15506	3899	49,67	74,85
T138M	2,81	7836	2421	25,10	69,10
M141F	3,5	9710	2790	31,10	71,27
C161F	2,73	3052	3209	9,78	-5,14
C161Y	2,59	4001	3582	12,82	10,47
C161S	2,17	10773	3904	34,51	63,76
C161G	2,41	12907	3750	41,34	70,95
C161R	2,75	5433	3987	17,40	26,62
R171N	1,06	9542	9888	30,57	-3,63
T169N	0,94	3798	2219	12,17	41,57
T169K	1,47	3320	2966	10,63	10,66
R171S	0,79	5774	4779	18,50	17,23
R171C	0,66	3365	2966	10,78	11,86
R171T	1,5	3320	2490	10,63	25,00
R171L	0,89	5774	4779	18,50	17,23
R171K	0,68	19001	8254	60,87	56,56
R171H	0,42	9179	7728	29,40	15,81
R171Q	0,53	17084	12930	54,72	24,32
R171G	0,5	11382	9938	36,46	12,69
R171I	0,5	26441	22335	84,70	15,53
R171M	0,99	17916	15952	57,39	10,96
R171W	0,6	11500	8408	36,84	26,89
R171P	0,96	10577	8010	33,88	24,27
W173L	2,05	102919	8618	329,68	91,63
W173S	2,68	4132	3527	13,24	14,64
W173G	0,94	15908	3288	50,96	79,33
S189C	2,18	5965	3663	19,11	38,59
S189L	1,38	9192	6898	29,44	24,96
S189W	3,09	8672	6991	27,78	19,38
S189A	1,32	8997	7074	28,82	21,37
W191R	1,84	5756	4154	18,44	27,83
W191G	1,47	8098	3745	25,94	53,75
W191S	1,03	5397	4772	17,29	11,58
W191C	1,34	6093	3705	19,52	39,19
W191L	2,32	4743	2375	15,19	49,93
V212M	1,27	5690	2480	18,23	56,41
FAT B Generation 1					
WT Mean	2,05	129284,5	9373,3	n.d.	91,5
A181V	1,63	85424,5	11994,5	66,07	85,96
V226C	0,57	162601,5	20143	125,77	87,61
M234T, I227Y	1,05	12217,5	12772,5	9,45	-4,54
A181V, V226C	0,51	102640	10711,5	79,39	89,56
A174G, A174G, V178A, A181V, G189A, V226C, I227Y, F281C	1,51	7040	6702	5,45	4,80

A174G, A174G, V178A, A181V, G189A, V226C, I227V, F281C	1,07	9372	8638	7,25	7,83
L166I, A102M, L175C, A174G, V178A, A181V, L184S, G185T, G189A, A190T, P192T, V203T, V226C, M234T, P270S, I227Y, P279V, F281C	0,74	7298,5	7643	5,65	-4,72
A174V	0,26	11277	11953	8,72	-5,99
A174D	0,8	9084	8971	7,03	1,24
L175F	0,63	8303,5	8998,5	6,42	-8,37
H177L	0,99	13338,5	7642,5	10,32	42,70
H177Q	1,07	8598,5	6180,5	6,65	21,35
H177D	0,95	7858,5	7894	6,08	-0,45
V178L	0,27	36588	33203,5	28,30	9,25
V178A	2,0	147853	10517	114,36	92,89
V178I	2,88	21593,5	6000,5	16,70	72,21
V178F	1,82	15751,5	15987	12,18	-1,50
V178M	1,25	15090,5	13658,5	11,67	9,49
V178G	1,4	248789	12700	192,44	94,90
A181E	1,53	43434,5	10751,5	33,60	75,25
A181D	1,65	9736,5	7745,5	7,53	20,45
L183M	1,44	42152	9553,5	32,60	77,34
L183Q	1,65	25283	9262	19,56	63,37
L183R	1,02	24966	14352	19,31	42,51
L183W	1,22	12056,5	9581	9,33	20,53
G189A	1,81	10582,5	8343	8,19	21,16
G189D	1,93	9844,5	10498	7,61	-6,64
G189C	0,74	15957,5	14771,5	12,34	7,43
G189V	1,75	8854,5	6325	6,85	28,57
G189E	2,36	10604	10640,5	8,20	-0,34
A190E	0,92	189859,5	12618	146,85	93,35
P192R	1,12	143533	10965,5	111,02	92,36
W201G	1,57	11133,5	11302	8,61	-1,51
W201L	1,52	11697	10806,5	9,05	7,61
FAT A Generation 2					
F118M	1,44	2697	2769	8,64	-3
F118G	1,18	2719,5	2902	8,71	-7
F118W	1,05	2898	2838,5	9,28	2
W136F	1,25	3285,5	3242,5	10,52	1
W136L	1,1	4620	4346	14,80	6
W136V	1,16	2383,5	2370	7,64	1
W136T	1,1	3796,5	3926	12,16	-3
W136C	1,33	3481,5	3162,5	11,15	9

W136R	0,81	4922	4971,5	15,77	-1
W136H	1,25	3120	3064,5	9,99	2
W136Q	1,37	3325	3302	10,65	1
W136I	1	3925,5	3579	12,57	9
W136N	1,16	3959	4043	12,68	-2
W136A	1,19	4287,5	4081,5	13,73	5
W136D	1,42	4253	4096	13,62	4
W136E	1,07	4128,5	3483,5	13,22	16
W136G	1,27	3788,5	3744,5	12,14	1
W136K	1,01	3367	3360,5	10,79	0
W136M	1,3	4362	4124	13,97	5
W136P	1,17	3627	3811,5	11,62	-5
W136S	1,3	3938,5	3733	12,62	5
W136Y	1,4	4161	3775	13,33	9
C161I	1,28	2384,5	2377,5	7,64	0
C161M	1,25	4611	4222	14,77	8
C161H	1,34	5621	5297	18,01	6
R171F	1,34	4326	4252	13,86	2
R171C	1,42	5020	4429	16,08	12
R171L	1,21	4423	3968	14,17	10
R171Q	1,51	5590,5	5342,5	17,91	4
R171S	1,31	3799,5	3077,5	12,17	19
R171T	1,39	3455,5	3235	11,07	6
R171N	1,35	3849	3479,5	12,33	10
R171H	1,11	3913,5	3086,5	12,54	21
R171G	1,38	2914,5	2813,3	9,34	-12
R171I	1,27	4748	3269	15,21	-18
R171K	1,14	3593,5	5582	11,51	2
R171M	1,19	3330,5	3509	10,67	1
R171W	1,05	4264,5	3283,5	13,66	5
R171V	0,9	5188	4067,5	16,62	-103
R171Y	0,93	5631,5	10548	18,04	-18
R171E	0,95	7874,5	6629	25,22	-10
R171D	0,9	7367	8638	23,60	-15
R171A	0,97	5242	8471	16,79	-7
R171P	0,5	5654,5	5584	18,11	-3
R186C	1,1	10050	5848	32,19	-14
S189G	1,1	1697,5	11437	5,44	20
S189V	1,32	1648	1363,5	5,28	12
S189K	1,56	2996	1448,5	9,60	14
S189I	1,33	2410,5	2569	7,72	15
W191F	1,7	2661	2046	8,52	-10
W191Y	1,39	2134	2924	6,84	6
W191L	1,26	2804	2003	8,98	-8
W191A	1	1795,5	3041	5,75	-11

W191C	1,13	2882,5	1994,5	9,23	-25
W191S	1,5	4078	3600	13,06	-7
W191G	1,48	1086,5	4351,5	3,48	-19
W191R	1,06	3990,5	1290,5	12,78	12
W191M	1,26	2586	3522,5	8,28	0
W191I	1,58	3783	2587	12,12	-14
W191V	1,56	4319,5	4319,5	13,84	10
W191H	1,85	2238	3866,5	7,17	22
W191T	1,15	2327,5	1756	7,46	21
W191K	1,17	1758	1829,5	5,63	7
W191N	1,28	1737	1637,5	5,56	0
W191P	1,69	1585	1732	5,08	-18
W191Q	0,58	1588	1871	5,09	16

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